

Format Converter Pro — Documentation

Overview

Format Converter Pro is a **browser-based, fully local** data file conversion tool. All processing happens entirely in your browser — no data is uploaded to any server. Files are read via JavaScript File APIs, processed locally, and downloaded directly to your device.

Table of Contents

- 1. Quick Start
- 2. Features
- 3. Step-by-Step Guide
- 4. Supported File Formats
- 5. Feature Deep Dives
- 6. Keyboard Shortcuts
- 7. Troubleshooting
- 8. Technical Architecture
- 9. Privacy & Security

Quick Start

- 1. Open `format-converter-pro.html` in any modern browser (Chrome, Firefox, Safari, Edge)
- 2. Click **“Start new conversion”** or select a **built-in template**
- 3. Upload a **target/sample file** with the desired output format
- 4. Upload your **input file(s)** to convert
- 5. Map columns, set transforms/filters, configure output settings
- 6. Validate, preview, and download

Features

Core Features

- **Rule-based column mapping** — Auto-map by similarity or manual selection
- **Format detection** — Automatically detects dates, numbers, text patterns from sample data
- **Multiple input files** — Batch process files with the same mapping
- **Profile saving** — Save conversion configurations for reuse

Pro Features (v4+)

#	Feature	Description
1	Data Validation	Pre-conversion quality checks: duplicates, type mismatches, blank values, date ranges
2	Transform Rules	Find/replace (regex), substring, conditional logic, math operations, prefix/suffix

Table 1 – continued

#	Feature	Description
3	Row Filtering	Include/exclude rows based on column conditions
4	Split Output	Split large outputs by row count or column value
5	Encoding Support	Auto-detect and convert between UTF-8, Shift_JIS, EUC-JP, UTF-16
6	Undo/Redo	50-step history with Ctrl+Z / Ctrl+Y
7	Template Library	Built-in templates + import/export profiles
8	Diff Preview	Side-by-side comparison of original vs converted data
9	Auto-Save Draft	Automatic recovery of interrupted sessions
10	Keyboard Shortcuts	Full keyboard navigation support

Step-by-Step Guide

Step 1: Start

Choose from:

-  **Start new conversion** — Begin from scratch
-  **Import a profile backup** — Load previously exported profiles JSON
- **Built-in templates** — Salesforce Contact, QuickBooks Journal, Shopify Product
- **Saved profiles** — Previously saved conversions in this browser

Step 2: Target Format

Upload a sample file representing your desired output format:

- Must include a **header row**
- Include **2+ data rows** for format detection
- Supported: CSV, TSV, TXT, JSON, XLSX, XLS

The app auto-detects:

- **Date formats** (YYYY-MM-DD, DD/MM/YYYY, Unix timestamps, etc.)
- **Number formats** (decimals, thousands separators, currency, negatives in parentheses)
- **Text patterns** (case, fixed length, padding)
- **Null representations** (N/A, NULL, blank, etc.)

Step 3: Input File(s)

Upload one or more files to convert:

- Files are **batched** together
- Select **input encoding** (auto-detect or manual: UTF-8, Shift_JIS, EUC-JP, UTF-16)
- Preview first 10 rows

Step 4: Column Mapping

For each target column, choose the source:

Action	Description
Map to input column	Link to a column from input file(s)
Map from sample record	Use a value from the target sample
Skip this column	Exclude from output
Blank	Output empty values
Default value	Use a fixed value for all rows

Auto-mapping uses Levenshtein similarity (60% match threshold).

Step 5: Transform & Filter

Transform Rules (applied per column, in order)

Transform	Use Case
Find & Replace	Clean data: “N/A” → “”, regex replacements
Substring	Extract codes: “INV-2024-001” → “2024” (start:4, length:4)
Conditional	If status=“A” then “Active” else “Inactive”
Math	Convert units: multiply by 1000, round, absolute value
Prefix/Suffix	Add currency symbols, IDs

Filter Rules (applied per row)

Operator	Example
equals	Status = “Active”
notEquals	Status ≠ “Deleted”
contains	Email contains “@company.com”
greaterThan	Amount > 1000
regex	SKU matches ¹ {3}-4\$/
empty / notEmpty	Check for blank values

Logic: All filter rules must pass (AND). Include rules require match; Exclude rules reject match.

Step 6: Output Settings

Format Options

- **Output format:** CSV, TSV, TXT, JSON, Pipe-delimited
- **Separator:** Comma, Pipe, Tab, Semicolon
- **Header row:** Include/exclude
- **Output encoding:** UTF-8, Shift_JIS, EUC-JP, UTF-16

¹ A-Z

Formatting Overrides

- Trim whitespace
- Skip empty rows
- Quote text values
- Override date format (e.g., DD/MM/YYYY)
- Override decimal places
- Override text case (UPPER, lower, Title)
- Replace empty values with custom string

Value Recoding Map specific input values to output values:

- “B” → “1” (Buy)
- “S” → “2” (Sell)
- “US” → “United States”

Split Output

- **By row count:** Max N rows per file (creates output_1.csv, output_2.csv, ...)
- **By column value:** Split by Region, Department, etc. (creates output_North.csv, output_South.csv, ...)

Step 7: Validation

Runs automated quality checks:

- **Duplicate rows** based on key columns
- **Blank values** per column (with percentage)
- **Type mismatches** (text in number columns, invalid dates)
- **Date range issues** (outside 1900-2100)

Results shown as:  Error |  Warning |  Info

Step 8: Preview

- **Standard view:** First 10 rows of converted output
- **Diff view** (toggle): Side-by-side original vs converted, with changed values highlighted

Step 9: Download

- Download converted file(s)
- Profile automatically saved to browser storage
- Options: Convert another file with same profile, start new conversion, or view saved profiles

Supported File Formats

Input Formats

Format	Extension	Notes
CSV	.csv	Comma-separated, with header Tab-separated Delimited text Array of objects [{"col": "val"}, ...]
TSV	.tsv	
TXT	.txt	
JSON	.json	
Excel	.xlsx, .xls	First sheet only

Output Formats

Format	Extension	Encoding Support
CSV	.csv	✔ All
TSV	.tsv	✔ All
TXT	.txt	✔ All
JSON	.json	UTF-8 only
Pipe-delimited	.txt	✔ All

Encoding Support

Encoding	Detection	Conversion
UTF-8	✔	✔
Shift_JIS (Japanese)	✔	✔
EUC-JP	✔	✔
UTF-16	✔	✔

Feature Deep Dives

Fixed-Length Text Padding (Issue Fix)

When the target format has fixed-length text fields (e.g., “silver” with 3 trailing spaces), the converter:

- 1. **Detects** the fixed length from sample data
- 2. **Pads** shorter values with trailing spaces to match target length
- 3. **Truncates** longer values (unless quote-text mode is enabled)

Date Format Handling

The converter recognizes and converts between:

- YYYY-MM-DD, DD/MM/YYYY, MM-DD-YYYY
- DD MMM YYYY (e.g., “15 JAN 2024”)
- Unix timestamps (seconds and milliseconds)
- Loose parsing for ambiguous formats

Custom formats use tokens: YYYY, YY, MM, M, DD, D, HH, mm, ss, A (AM/PM)

Number Format Handling

Detects and preserves:

- Decimal places
- Thousands separators
- Currency symbols (\$, €, £,)
- Negative numbers in parentheses
- Zero-padding

Keyboard Shortcuts

Shortcut	Action
Ctrl + Enter	Next step
Ctrl + Shift + Enter	Run validation / confirm download
Ctrl + Z	Undo
Ctrl + Y	Redo
Esc	Go back

Troubleshooting

Common Issues

“No columns detected” → Ensure the first row contains column headers, not data.

“Type mismatch errors in validation” → Check that mapped columns contain the expected data type. Use transform rules to clean data before conversion.

“Special characters corrupted” → Try manually selecting the input encoding (Step 3) instead of auto-detect.

“Scroll jumps when selecting in Step 4” → Fixed in v4+. If persists, try using keyboard (Tab/Arrow keys) instead of mouse.

“Profile not showing after restart” → Profiles are stored in browser localStorage. They persist per browser but don’t sync across devices. Use Export/Import to transfer.

“Draft recovery prompt keeps appearing” → Click “Discard draft” or clear browser localStorage for this site.

Browser Compatibility

-  Chrome 80+
-  Firefox 75+
-  Safari 13.1+
-  Edge 80+

Requires JavaScript enabled. No browser extensions needed.

Technical Architecture

Data Flow

```
User Files → FileReader API → PapaParse/XLSX → JavaScript Objects →  
Transform Pipeline → Validation → Formatting → Output Generator →  
Blob → Download
```

Storage

Data	Location	Persistence
Saved profiles	localStorage	Until manually cleared
Draft recovery	localStorage	7 days max
Undo history	Memory (RAM)	Lost on page reload

Libraries Used

- **SheetJS (xlsx)** — Excel file parsing
- **PapaParse** — CSV/TSV parsing
- **encoding-japanese** — Character encoding detection/conversion

No Server Required

The entire application is a single HTML file (~94 KB) that runs client-side. No build step, no dependencies to install, no backend API.

Privacy & Security

- **100% local processing** — Files never leave your browser
- **No external API calls** — Except CDN libraries (loaded once)
- **No tracking** — No analytics, cookies, or telemetry
- **Storage is local** — Data saved only in your browser's localStorage
- **Share carefully** — Exported profiles JSON may contain sample data snippets

Version History

Version	Date	Changes
v1	—	Initial release
v2	—	Bug fixes
v3	—	Format detection improvements
v4	—	Scroll fix, fixed-length padding

Table 10 – continued

Version	Date	Changes
v5 (Pro)	2026	All 10 features: validation, transforms, filters, split output, encoding, undo/redo, templates, diff preview, auto-save, keyboard shortcuts

License

MIT License — Free for personal and commercial use.

Generated for Format Converter Pro v5